Chapter 5: Case Study – Measles in Large and Small Towns

Source code at pypomp/tutorials/sbied/chapter5

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Objectives

- To display a published case study using plug-and-play methods with non-trivial model complexities.
- To show how extra-demographic stochasticity can be modeled.
- To demonstrate the use of covariates in `pypomp`.
- To demonstrate the use of profile likelihood in scientific inference.
- To discuss the interpretation of parameter estimates.
- To emphasize the potential need for extra sources of stochasticity in modeling.

Challenges in inference from disease dynamics

Understanding, forecasting, and managing epidemiological systems increasingly depends on models. Dynamic models can be used to test causal hypotheses.

Real epidemiological systems:

- are nonlinear
- are stochastic
- are nonstationary
- evolve in continuous time
- have hidden variables
- can be measured only with (large) error

Measles is the paradigm for a nonlinear ecological system that can be well described by low-dimensional nonlinear dynamics.

A tradition of careful modeling studies have proposed and found evidence for a number of specific mechanisms, including

- a high value of $\mathcal{R}_0$ (c. 15–20)
- under-reporting
- seasonality in transmission rates associated with school terms
- response to changing birth rates
- a birth-cohort effect
- metapopulation dynamics
- fadeouts and reintroductions that scale with city size

- spatial traveling waves

Much of this evidence has been amassed from fitting models to data, using a variety of methods. See Rohani and King (2010) for a review of some of the high points.

Measles in England and Wales

We revisit a classic measles data set, weekly case reports in 954 urban centers in England and Wales during the pre-vaccine era (1950–1963).

We examine questions regarding:

- measles extinction and recolonization
- transmission rates
- seasonality
- resupply of susceptibles

We use a model that

1. expresses our current understanding of measles dynamics
2. includes a long list of mechanisms that have been proposed and demonstrated in the literature
3. cannot be fit by previous likelihood-based methods

We examine data from large and small towns using the same model, something no existing methods have been able to do.

We ask: does our perspective on this disease change when we expect the models to explain the data in detail? What bigger lessons can we learn regarding inference for dynamical systems?

Data sets

He et al. (2010) studied twenty towns, including

- 10 largest cities in England and Wales
- 10 smaller towns, chosen at random

Population sizes ranged from about 2,000 to 3.4 million. Weekly case reports span 1950–1963, with annual birth records and population sizes from 1944–1963.

```
all_data = UKMeasles.subset()
measles = all_data["measles"]
demog = all_data["demog"]
```

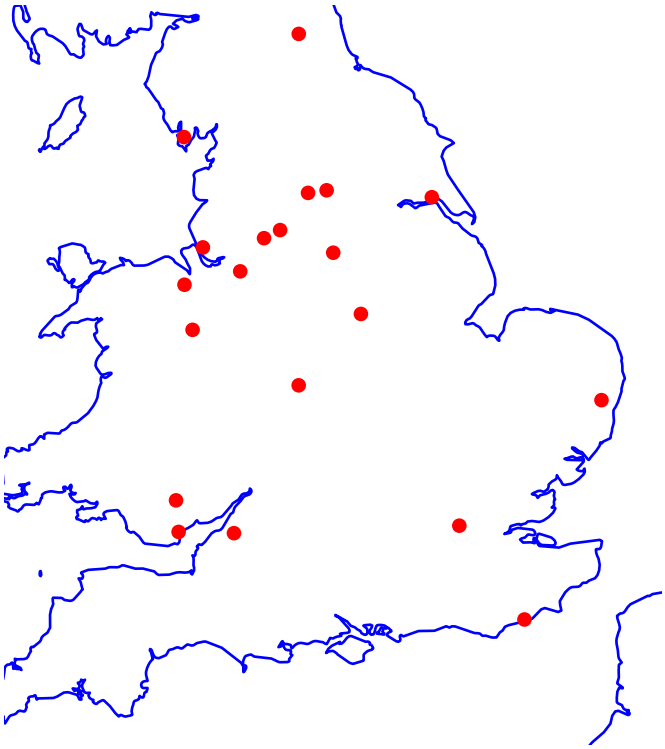
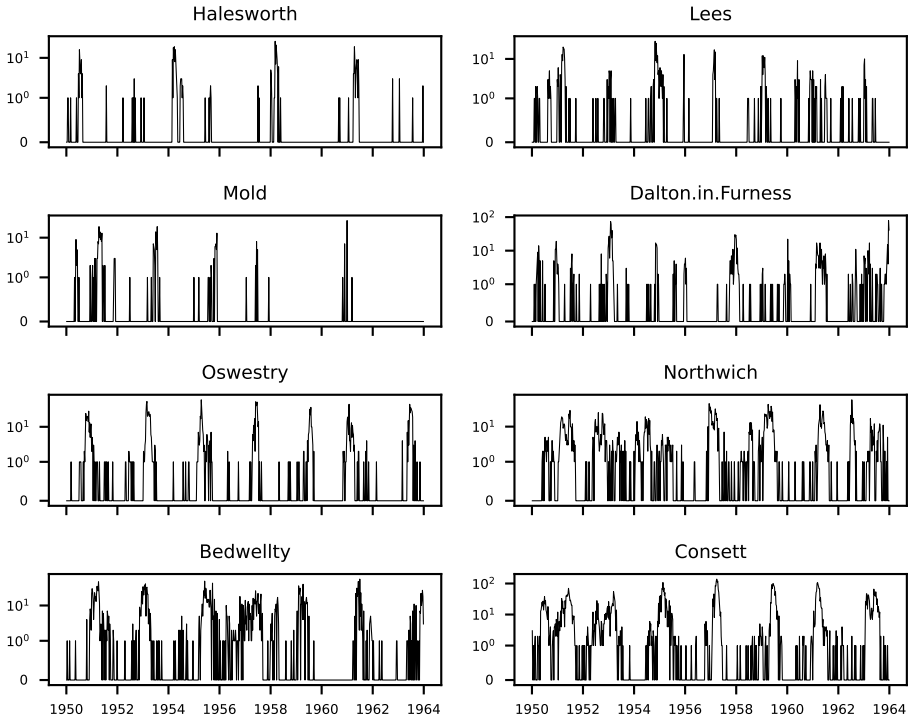


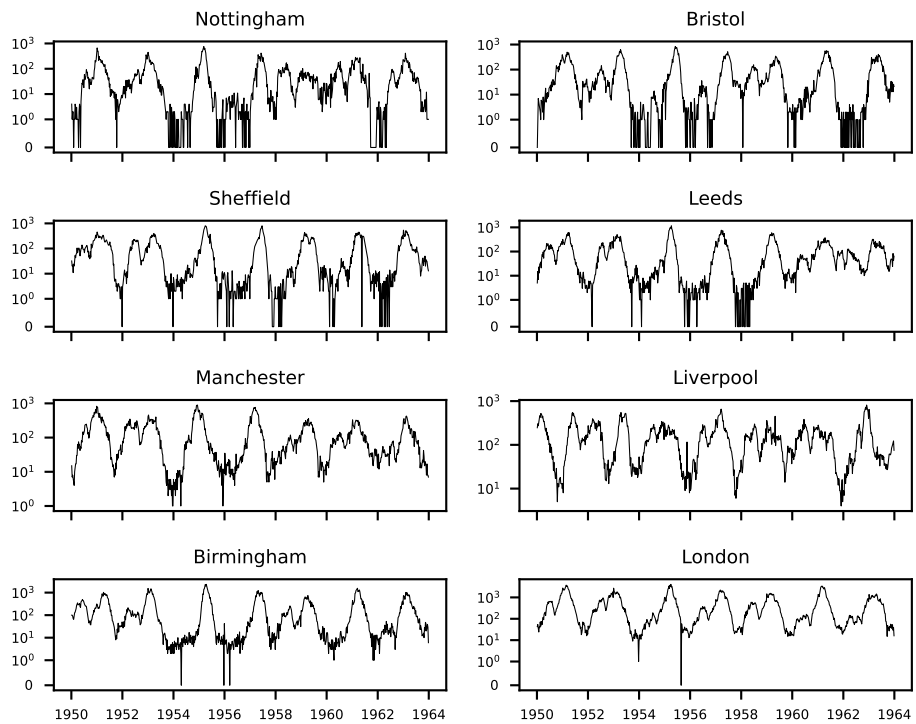
Figure 1: Locations of the 20 cities in the He et al. (2010) analysis.

Map of cities in the analysis

City case counts: smallest 8 cities

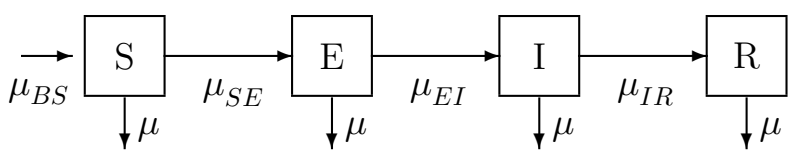


City case counts: largest 8 cities



Continuous-time Markov process model

The model is a SEIR (susceptible–exposed–infectious–recovered) compartment model with births and deaths:



Model specification

Covariates (from data):

- $B(t)$: birth rate
- $N(t)$: population size

Entry into susceptible class (birth-cohort effect):

$$\mu_{BS}(t) = (1-c)\,B(t-\tau) + c\,\delta(t - \lfloor t \rfloor) \int_{t-1}^{t} B(t-\tau-s)\,ds$$

- c: cohort effect (fraction entering at school start)
- τ: school-entry delay
- $\lfloor t \rfloor$: most recent September~1 before t

Force of infection:

$$\mu_{SE}(t) = \frac{\beta(t)}{N(t)}\,(I+\iota)\,\zeta(t)$$

- ι: imported infections
- $\zeta(t)$: Gamma white noise with intensity σ_{SE} (He et al., 2010; Bhadra et al., 2011)

School-term transmission:

$$\beta(t) = \begin{cases} \beta_0\,(1+a(1-p)/p) & \text{during term} \\ \beta_0\,(1-a) & \text{during vacation} \end{cases}$$

- a: amplitude of seasonality
- $p = 0.7589$: fraction of the year children are in school
- The factor $(1-p)/p$ ensures the average transmission rate is β_0.

Overdispersed measurement model:

$$\text{cases}_t \,\big|\, \Delta N_{IR} = C_t \sim \text{Normal}\big(\rho\,C_t,\ \rho\,(1-\rho)\,C_t + (\psi\,\rho\,C_t)^2\big)$$

- ρ: reporting rate
- ψ: overdispersion parameter
- When $\psi = 0$, the variance–mean relation reduces to that of the binomial distribution.

The partially observed Markov process model

We require a simulator for our model. Notable complexities include:

1. Incorporation of the known birthrate.
2. The birth-cohort effect: a specified fraction of the cohort enters the susceptible pool all at once.
3. Seasonality in the transmission rate: higher during school terms than holidays.
4. Extra-demographic stochasticity in the form of a Gamma white-noise term acting multiplicatively on the force of infection.
5. Demographic stochasticity implemented using Euler-multinomial distributions.

Implementation of the process model

The process model is implemented in `pypomp.measles.model_001b`. Let's walk through it.

State variables and parameters:

```
statenames = ["S", "E", "I", "R", "W", "C"]
accumvars = ["W", "C"]

param_names = (
    "R0", "sigma", "gamma", "iota", "rho",
    "sigmaSE", "psi", "cohort", "amplitude",
    "S_0", "E_0", "I_0", "R_0",
)
```

- C is the true incidence (accumulator for new infections), reset each observation interval.
- W is cumulative white noise, useful for diagnostics.

Cohort effect: at the start of the school year (day~251), a fraction `cohort` of the year's births enter susceptibles all at once:

```
t_mod = t - jnp.floor(t)
is_cohort_time = jnp.abs(t_mod - 251.0/365.0) < 0.5*dt
br = jnp.where(
    is_cohort_time,
    cohort * birthrate / dt + (1 - cohort) * birthrate,
    (1 - cohort) * birthrate,
)
```

Term-time seasonality:

```
t_days = t_mod * 365.25
in_term_time = (
    ((t_days >= 7) & (t_days <= 100))
    | ((t_days >= 115) & (t_days <= 199))
    | ((t_days >= 252) & (t_days <= 300))
    | ((t_days >= 308) & (t_days <= 356))
)
seas = jnp.where(
    in_term_time,
    1.0 + amplitude * 0.2411 / 0.7589,
    1 - amplitude
)
beta = R0 * seas * (1.0 - jnp.exp(-(gamma + mu)*dt)) / dt
```

Extra-demographic stochasticity and transitions:

```
# Force of infection
foi = beta * (I + iota) / pop

# Gamma white noise
dw = fast_approx_rgamma(keys[0], dt/sigmaSE**2) * sigmaSE**2

# Rates for Euler-multinomial transitions
rate = jnp.array([foi*dw/dt, mu, sigma, mu, gamma, mu])

# Poisson births
births = fast_approx_rpoisson(keys[1], br * dt)

# Euler-multinomial transitions
transitions = fast_approx_rmultinom(keys[2], populations, rt_final)
```

- `fast_approx_rgamma`, `fast_approx_rpoisson`, and `fast_approx_rmultinom` are JAX-compatible random variate generators provided by `pypomp`.

State update:

```
S = S + births - trans_S[0] - trans_S[1]
E = E + trans_S[0] - trans_E[0] - trans_E[1]
I = I + trans_E[0] - trans_I[0] - trans_I[1]
R = pop - S - E - I
W = W + (dw - dt) / sigmaSE
C = C + trans_I[0]
```

- Since recognized measles cases are quarantined, most infection occurs before case recognition.

True incidence C counts individuals progressing from I to R.

State initializations

The initial state allocates the population across compartments according to proportions S_0, E_0, I_0, R_0:

```
def rinit(theta_, key, covars, t0):
    S_0, E_0, I_0, R_0 = (
        theta_["S_0"], theta_["E_0"],
        theta_["I_0"], theta_["R_0"])
    m = covars["pop"] / (S_0 + E_0 + I_0 + R_0)
    S = jnp.round(m * S_0)
    E = jnp.round(m * E_0)
    I = jnp.round(m * I_0)
    R = jnp.round(m * R_0)
    return {"S": S, "E": E, "I": I, "R": R,
        "W": 0, "C": 0}
```

Measurement model

We model both under-reporting and measurement error. We want $\mathbb{E}[\text{cases}|C] = \rho\, C$ and $\text{Var}[\text{cases}|C] = \rho\,(1-\rho)\,C + (\psi\,\rho\,C)^2$.

Specifically, cases $\mid C \sim f(\cdot \mid \rho, \psi, C)$, where

$$\begin{aligned} f(c \mid \rho, \psi, C) \quad = \quad & \Phi\big(c + \tfrac{1}{2},\, \rho C,\, \rho(1-\rho)C + (\psi\rho C)^2\big) \\ & -\Phi\big(c - \tfrac{1}{2},\, \rho C,\, \rho(1-\rho)C + (\psi\rho C)^2\big)\,. \end{aligned}$$

Here, $\Phi(x, \mu, \sigma^2)$ is the c.d.f. of the normal distribution with mean μ and variance σ^2.

The measurement density `dmeas`:

```
def dmeas(Y_, X_, theta_, covars, t):
    rho, psi, C = theta_["rho"], theta_["psi"], X_["C"]
    m = rho * C
    v = m * (1.0 - rho + psi**2 * m)
    sqrt_v = jnp.sqrt(v) + 1e-18
    y = Y_["cases"]
    upper = jax.scipy.stats.norm.cdf(y + 0.5, m, sqrt_v)
    lower = jax.scipy.stats.norm.cdf(y - 0.5, m, sqrt_v)
    lik = jnp.where(y > 1e-18, upper - lower, upper) + 1e-18
    return jnp.log(lik)
```

The measurement simulator `rmeas`:

```
def rmeas(X_, theta_, key, covars, t):
    rho, psi, C = theta_["rho"], theta_["psi"], X_["C"]
    m = rho * C
    v = m * (1.0 - rho + psi**2 * m)
    cases = jax.random.normal(key) * (jnp.sqrt(v) + 1e-18) + m
    return jnp.where(cases > 0.0, jnp.round(cases), 0.0)
```

Data and covariates

The `pypomp` package includes the UK measles data from He et al. (2010). The `UKMeasles` class provides convenient access.

We illustrate using London:

```
london_data = UKMeasles.subset(units=["London"])
dat = london_data["measles"]
print(f"Date range: {dat['date'].min()}",
    f" to {dat['date'].max()}")
print(f"Number of observations: {len(dat)}")
```

```
Date range: 1944-01-07 00:00:00  to 1965-03-26 00:00:00
Number of observations: 1108
```

The case report data

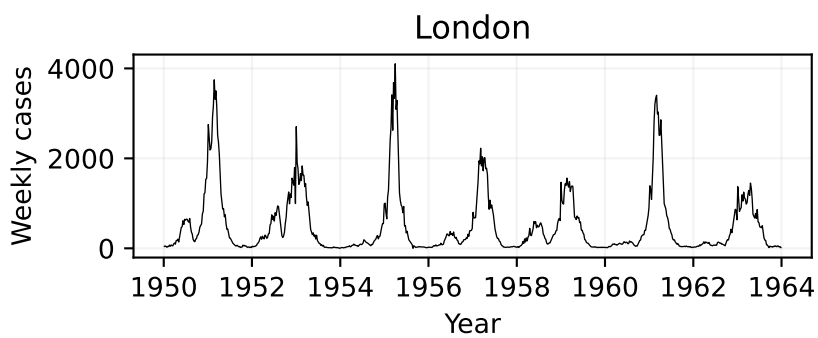


Figure 2: Weekly measles case reports, London 1950–1963.

Reviewing covariates in time series analysis

Suppose our time series of primary interest is $y_{1:N}$. A **covariate time series** is data $z_{1:N}$ which is used to help explain $y_{1:N}$.

- It may be implicit that a covariate $z_{1:N}$ is considered an **external forcing** to the system producing $y_{1:N}$, i.e., $z_{1:N}$ causally affects $y_{1:N}$ but not vice versa.
- E.g., weather might affect human health, but human health has negligible effect on weather: weather is an external forcing to human health.

When the process leading to $z_{1:N}$ is not external to the system generating it, we must be alert to the possibility of **reverse causation** and **confounding variables**.

- Here, births are external, assuming negligible effect of measles on live births.
- For a mechanistic POMP, a covariate does not have to enter the equations linearly, as in a standard linear model.

Covariates in Pypomp

- A covariate table is incorporated into the POMP object by the `covars` argument to `pp.Pomp`.

- The named coviariates are delivered to the model components, after look up at the corresponding time, `t`.

Other common POMP covariates

A linear or nonlinear trend through time in any parameter or rate or latent state can be modeled using covariates.

- Seasonality can be modeled as a covariate which can enter the model in any scientifically plausible way.
- Periodic B-splines provide a flexible way to incorporate seasonality; the source code for `pp.dacca` provides an example.
- The measles model describes seasonality directly via the `t` variable accessible in all the model components.

Covariate pre-processing for measles

- Discretely measured covariates must be interpolated at arbitrary times for a continuous-time POMP.

We interpolate the covariates and delay the entry of newborns into the susceptible pool. This is handled internally by `UKMeasles.Pomp()`, which uses smoothing splines for population interpolation and shifts the birth rate by 4 years (the school-entry delay τ).

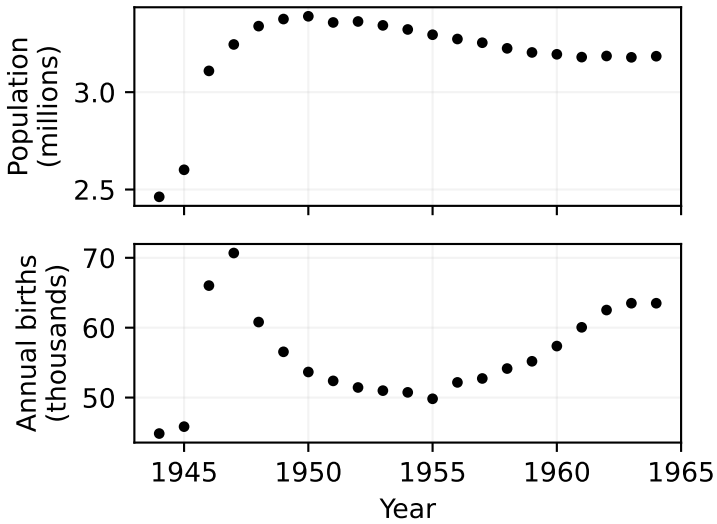


Figure 3: Population and birth-rate covariates for London.

Constructing the POMP object

```
theta_london = mles["London"].to_dict()
```

```
m1 = UKMeasles.Pomp(
    unit=["London"],
    theta=theta_london,
    model="001b",
    interp_method="shifted_splines",
    first_year=1950,
    last_year=1963,
    dt=1/365.25,
    clean=True
)
```

```
State names: ['S', 'E', 'I', 'R', 'W', 'C']
Time range: 1950.01 to 1963.98
Observations: 730
```

Estimates from He et al. (2010)

He et al. (2010) estimated the parameters of this model for all 20 cities. We verify that we get the same likelihood.

```
key = jax.random.key(998468235)
cache_file = cache_dir + "/london_pf.pkl"
if os.path.exists(cache_file):
    with open(cache_file, 'rb') as f:
        london_pf = pickle.load(f)
else:
    m1.pfilter(J=[50,500,5000][RL],
        reps=[2,5,10][RL], key=key, thresh=0,
        CLL=True, ESS=True)
    london_pf = m1.results_history[-1]
    with open(cache_file, 'wb') as f:
        pickle.dump(london_pf,f)

london_logliks = london_pf.logLiks.values.flatten()
ll = pp.logmeanexp(london_logliks)
ll_se = pp.logmeanexp_se(london_logliks)
print(f"Log-likelihood: {ll:.1f} (SE {ll_se:.2f})")
```

```
Log-likelihood: -3806.1 (SE 0.25)
```

Simulations at the MLE

```
key = jax.random.key(42)
X_sims, Y_sims = m1.simulate(key=key, nsim=3)
```

Parameter transformations

The parameters are constrained to be positive, and some lie between 0 and 1. We turn the likelihood maximization problem into an unconstrained one by transforming:

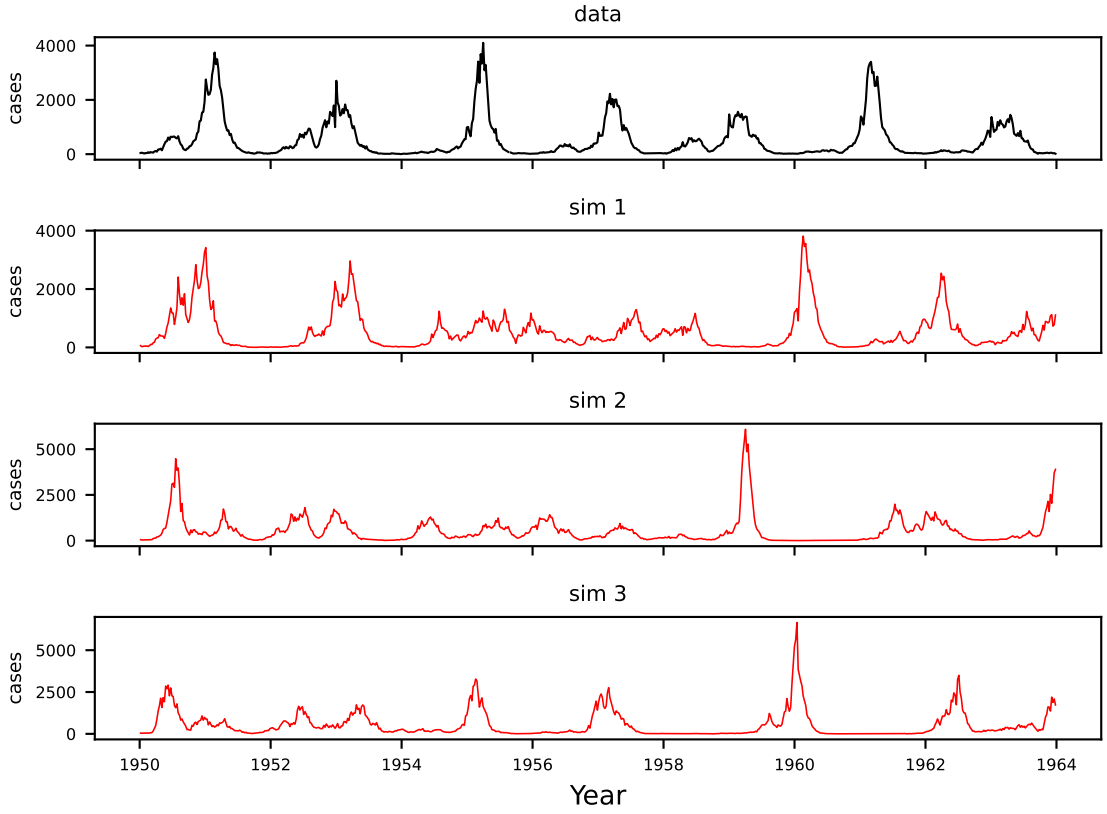


Figure 4: Simulations (red) versus data (black) at the London MLE.

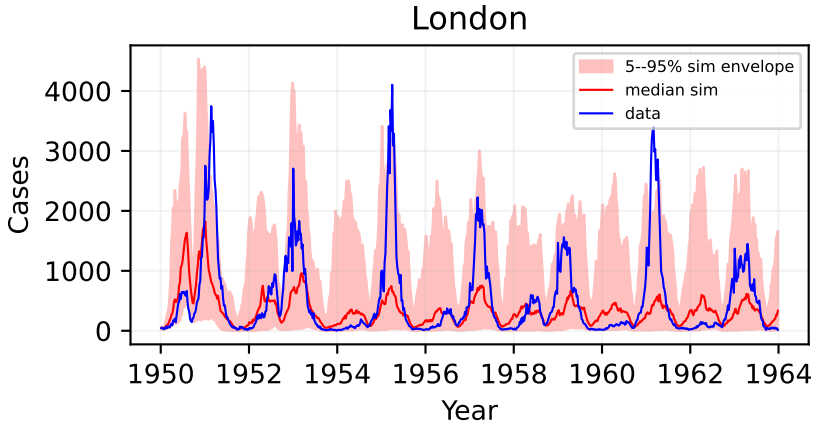


Figure 5: Simulation envelope (100 sims) versus data.

- **Positive parameters** (R_0, σ, γ, ι, σ_{SE}, ψ): log transform
- **Parameters in** $(0,1)$ (ρ, cohort, amplitude): logit transform
- **Initial proportions** (S_0, E_0, I_0, R_0): log-barycentric transform

These are implemented in `model_001b.to_est` and `model_001b.from_est`.

ARMA benchmark

Linear, Gaussian auto-regressive moving-average (ARMA) models provide a flexible non-mechanistic benchmark.

```
from statsmodels.tsa.arima.model import ARIMA

cases = m1.ys["cases"].values
log_y = np.log(np.maximum(cases, 1))

arma_fit = ARIMA(log_y, order=(2, 0, 2)).fit()
arma_loglik = arma_fit.llf - np.sum(log_y)

print(f"ARMA(2,2) log-lik: {arma_loglik:.1f} (p=5)")
print(f"SEIR model log-lik: {ll:.1f} (p=12)")
```

```
ARMA(2,2) log-lik: nan (p=5)
SEIR model log-lik: -3806.1 (p=12)
```

- Minimizing AIC, $2p - 2\ell$, is equivalent to maximizing $\ell - p$.
- The aim of mechanistic modeling is not to beat benchmarks, but falling far behind can diagnose problems.
- "Far" means many log units: differences of log-likelihoods are invariant to the scale of measurement.

Log-likelihood anomalies

The benchmark and model log-likelihoods can both be decomposed as a sum of conditional log-likelihoods,

$$\ell(\theta) = \sum_{n=1}^{N} \ell_n(\theta) \quad \text{where} \quad \ell_n(\theta) = \log f_{Y_n|Y_{1:n-1}}(y_n^* \,|\, y_{1:n-1}^*;\, \theta).$$

The **anomaly** for the model at time t_n is the difference between the model conditional log-likelihood and that of the benchmark.

Anomalies can be used similarly to regression residuals: they can indicate points where the model fails; patterns can reveal scope for model improvement.

Particle filter variance

Effective sample size is $\text{ESS} = \left(\sum_{j=1}^{J} w_j^2\right)^{-1}$, where w_j are the resampling probabilities, i.e., the normalized filter weights.

- ESS is the equivalent number of independent particles.

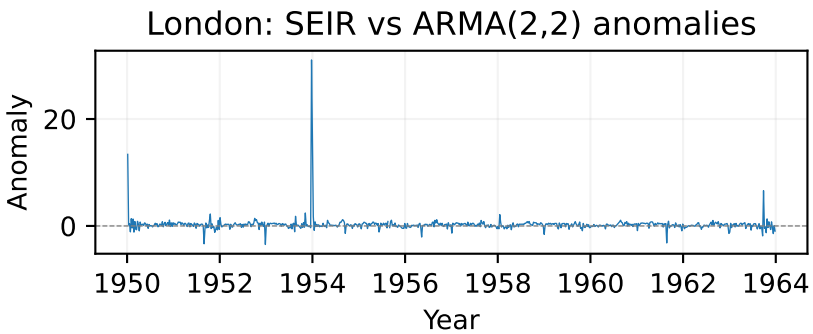


Figure 6: Conditional log-likelihood anomalies for London. Positive anomalies indicate the model outperforms ARMA; negative anomalies indicate the model struggles.

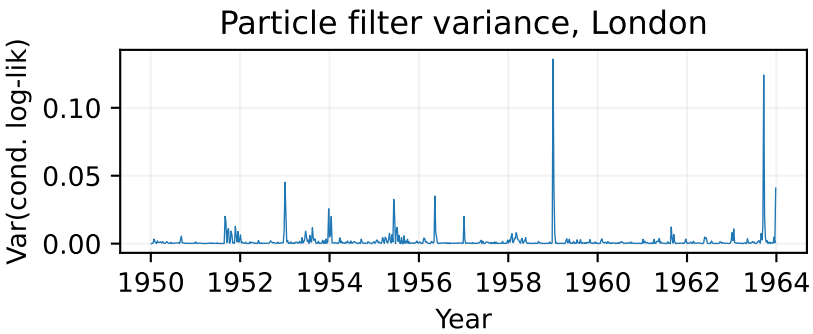


Figure 7: Variance of conditional log-likelihoods across replicates. Observations with high variance are numerically problematic.

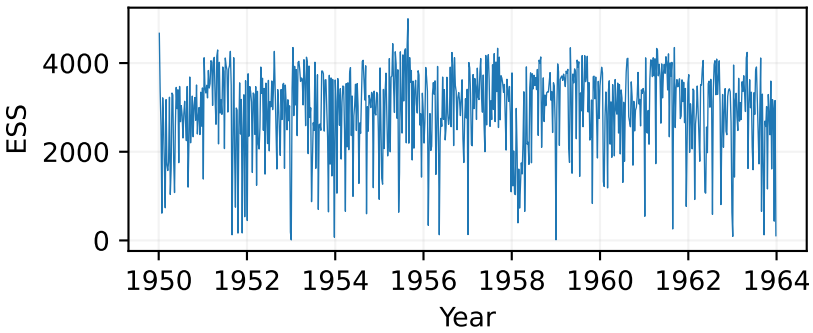


Figure 8: Effective sample size for one particle filter in London.

- If $w_j = 1/J$ for all j then ESS $= J$.

Interpretation of ESS

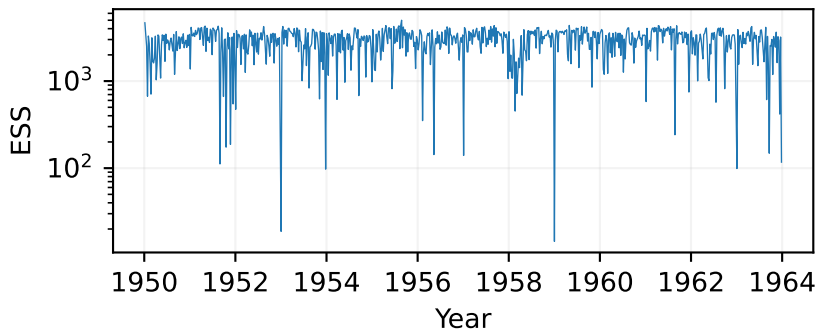


Figure 9: Mean of replicated ESS.

- As a rule of thumb, ESS < 100 is not good.
- Some time points here are problematic and could usefully be investigated further. Is there a problem with either the model or the data at these points?

Results from He et al. (2010)

The fitting procedure used is as follows:

- A large number of searches were started at points across the parameter space.
- Iterated filtering was used to maximize the likelihood.
- Point estimates of all parameters were obtained for 20 cities.
- Profile likelihoods were constructed to quantify uncertainty in London and Hastings.

The linked document `lesson5_profile.qmd` shows how a likelihood profile can be constructed using IF2 in `pypomp`.

Setting up IF2

The IF2 algorithm requires specifying random walk standard deviations for each parameter being estimated:

```
from pypomp import RWSigma

rw_sd = RWSigma(
    sigmas={
        "R0": 0.02, "sigma": 0.02, "gamma": 0.02,
        "iota": 0.0, "rho": 0.0,
        "sigmaSE": 0.02, "psi": 0.02,
        "cohort": 0.02, "amplitude": 0.02,
        "S_0": 0.02, "E_0": 0.02, "I_0": 0.02, "R_0": 0.02
    },
    init_names=["S_0", "E_0", "I_0", "R_0"]
)
```

```
m1.mif(J=2000, M=50, a=0.95, rw_sd=rw_sd,
       key=jax.random.key(1234567), thresh=0)
```

town	pop	R0	a	LP	IP	iota	rho	psi	sigSE
Halesworth	2171	33.1	0.38	7.9	2.3	0.01	0.754	0.64	0.075
Lees	4247	29.7	0.15	8.5	2.1	0.03	0.612	0.68	0.080
Mold	6409	21.4	0.27	5.9	1.8	0.01	0.131	2.87	0.054
Dalton.in.Furness	10560	28.3	0.20	5.5	2.0	0.04	0.455	0.82	0.078
Oswestry	10970	52.9	0.34	10.3	2.7	0.03	0.631	0.48	0.070
Northwich	18330	30.1	0.42	8.5	3.0	0.06	0.795	0.40	0.086
Bedwellty	28930	24.7	0.16	6.8	3.0	0.04	0.311	0.95	0.061
Consett	39130	35.9	0.20	9.1	2.7	0.07	0.650	0.41	0.071
Hastings	65690	34.2	0.30	7.0	5.4	0.19	0.695	0.40	0.096
Cardiff	244600	34.4	0.22	9.9	3.1	0.14	0.602	0.27	0.054
Bradford	294300	32.1	0.24	8.5	3.4	0.24	0.599	0.19	0.045
Hull	302100	38.9	0.22	9.2	5.5	0.14	0.582	0.26	0.064
Nottingham	307000	22.6	0.16	5.7	3.7	0.17	0.609	0.26	0.038
Bristol	442600	26.8	0.20	6.2	4.9	0.44	0.626	0.20	0.039
Leeds	509700	47.8	0.27	9.5	10.9	1.25	0.666	0.17	0.078
Sheffield	515000	33.1	0.31	7.2	6.4	0.85	0.649	0.18	0.043
Manchester	704500	32.9	0.29	11.1	6.9	0.59	0.550	0.16	0.055
Liverpool	802300	48.1	0.30	7.9	9.8	0.26	0.494	0.14	0.053
Birmingham	1117900	43.4	0.43	8.5	11.6	0.34	0.544	0.18	0.061
London	3389620	56.8	0.55	13.1	12.5	2.90	0.488	0.12	0.088

Some parameter estimates vary with city size

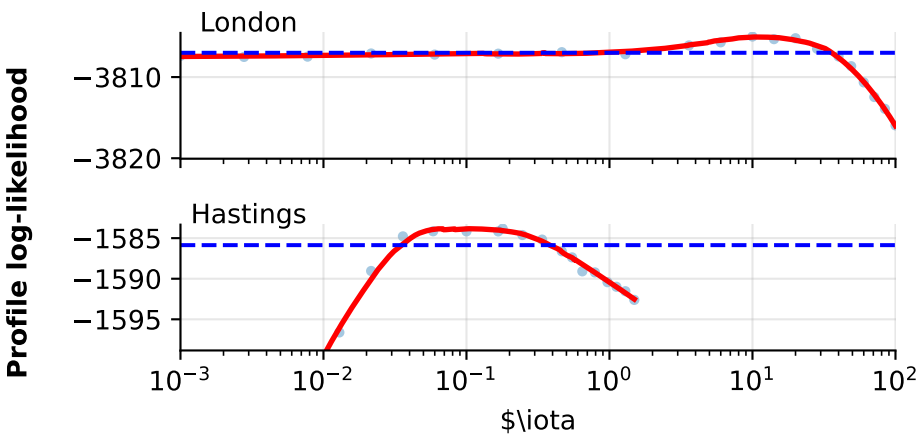
The Spearman rank correlations between each parameter and N_{1950} reveal which parameters vary systematically with city size.

```
Spearman r(parameter, N_1950):
  R0          : +0.46
  a           : +0.31
  LP          : +0.31
  IP          : +0.95
  iota        : +0.93
  rho         : -0.20
  psi         : -0.93
  sigSE       : -0.31
```

Imported infections

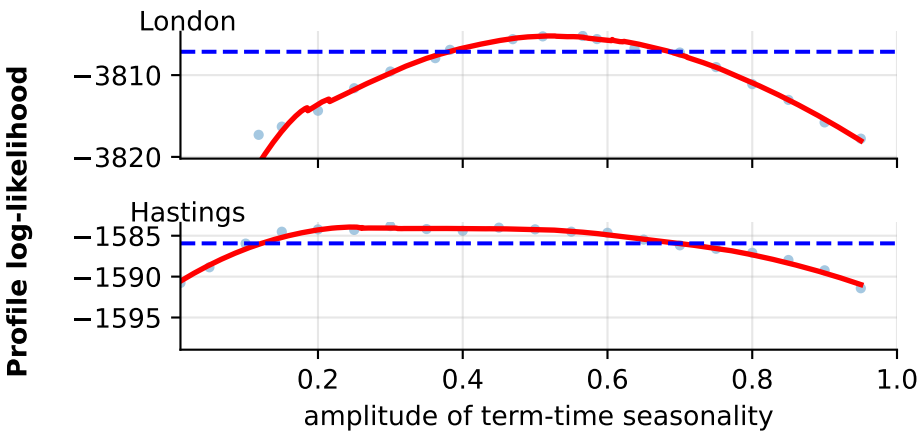
For implementation, see the exercise notes and their source code.

$$\text{force of infection} = \mu_{SE} = \frac{\beta(t)}{N(t)}\,(I+\iota)\,\zeta(t)$$



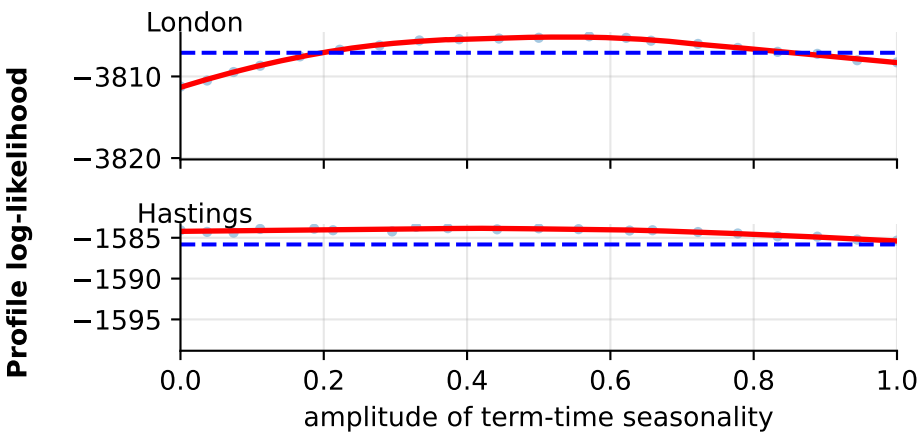
Seasonality

- Both London and Hastings have high seasonality amplitude ($a > 0.4$).
- School terms drive measles transmission strongly.
- This parameter is well-identified, especially in London.



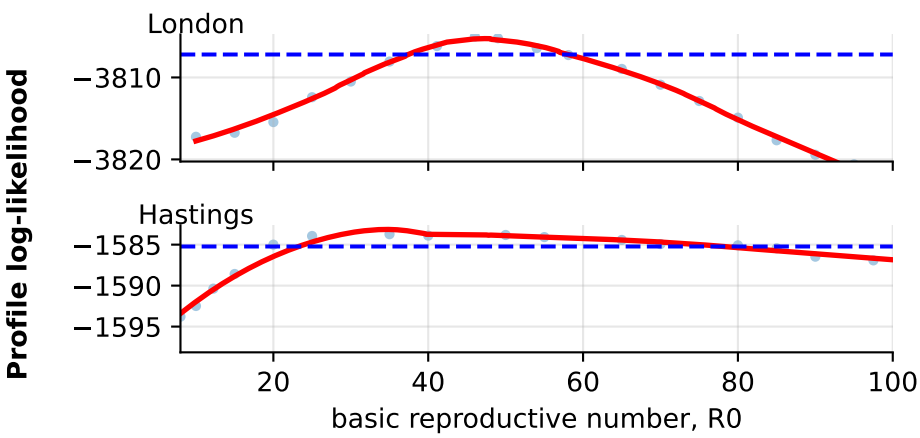
Cohort effect

- The cohort effect is moderately well-identified in London.
- The flatter profile in Hastings suggests weaker identifiability in small populations.



R_0 estimates inconsistent with literature

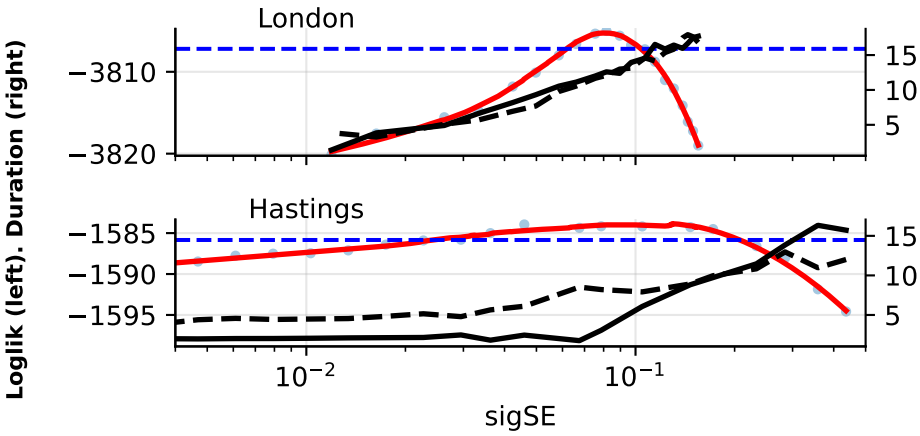
Recall that $\mathcal{R}_0$ is a measure of how communicable an infection is. Existing estimates ($\mathcal{R}_0 \approx 15$–20) come from serology surveys and feature-based methods.



Extrademographic stochasticity

$$\mu_{SE} = \frac{\beta(t)}{N(t)}\,(I+\iota)\,\zeta(t)$$

$\sigma_{SE} > 0$ is strongly supported by the data.
Trace plots show trade-offs with the infectious period (IP, dashed black line) and latent period (LP, solid black line).



Reporting rate

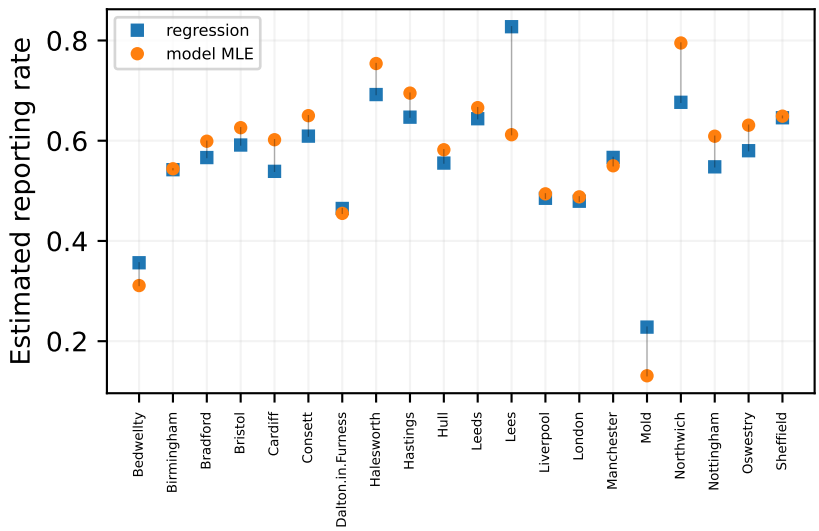


Figure 10: Reporting rate: regression estimate (cases/births slope) versus model MLE.

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Predicted vs observed critical community size

Questions

- What does it mean that parameter estimates from the fitting disagree with estimates from other data?
- How can one interpret the correlation between infectious period and city size in the parameter estimates?
- How do we interpret the need for extrademographic stochasticity in this model?

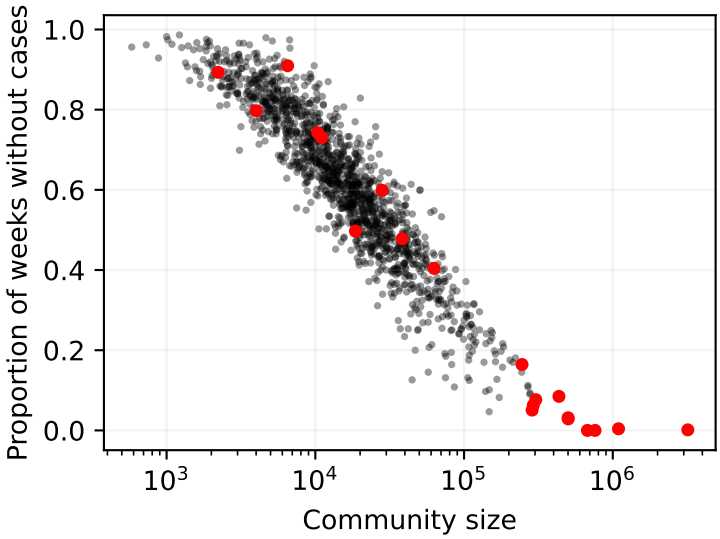

Figure 11: Fraction of weeks with zero cases versus population size. The critical community size is approximately 300,000.

Exercise 5.1. Reformulate the model

- Modify the He et al. (2010) model to remove the cohort effect. Run simulations and compute likelihoods to convince yourself that the resulting codes agree with the original ones for `cohort = 0`.
- Now modify the transmission seasonality to use a sinusoidal form. How many parameters must you use? Fixing the other parameters at their MLE values, compute and visualize a profile likelihood over these parameters.

Exercise 5.2. Extrademographic stochasticity

Set the extrademographic stochasticity parameter $\sigma_{SE} = 0$, set $\alpha = 1$ (already the case in `model_001b`), and fix ρ and ι at their MLE values, then maximize the likelihood over the remaining parameters.

- How do your results compare with those at the MLE? Compare likelihoods but also use simulations to diagnose differences between the models.

References

Bhadra, A., Ionides, E. L., Laneri, K., Pascual, M., Bouma, M., and Dhiman, R. (2011). Malaria in northwest India: data analysis via partially observed stochastic differential equation models driven by Lévy noise. *J Am Stat Assoc*, 106(494):440–451.

He, D., Ionides, E. L., and King, A. A. (2010). Plug-and-play inference for disease dynamics: measles in large and small populations as a case study. *J R Soc Interface*, 7:271–283.

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